DATA-DRIVEN PREDICTION AND ANALYSIS OF COOLING ENERGY USE IN TELECOM CENTRAL OFFICE ENVIRONMENTS

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Outline

- Telecom cooling problem and telemetry gap
- Aim, tasks, and research questions
- Benchmark-based control evaluation framework
- From fragmented telemetry to cooling-demand target
- Expected-demand forecasting and rolling validation
- Lagged demand, load proxy, and thermal context
- Control strategies and historical impact
- Interpretation, limitations, and thesis conclusions

Problem and motivation

- Telecom central offices require continuous cooling for reliable operation of sensitive equipment
- Existing telecom analytics are stronger in monitoring and diagnosis than in control-oriented decision support
- More intelligent cooling control could improve efficiency without compromising operational safety, but full advanced control is difficult under available telemetry constraints
- The thesis therefore develops an intelligent, control-oriented framework for real telecom operating conditions.

Research gap and literature background

- Telecom studies are relatively strong in monitoring, KPI design, and diagnosis
- Real cooling telemetry is often not directly suitable for control-oriented analysis and may require reconstruction of a defensible analytical basis
- Literature provide useful methods, but often assume ready targets, richer state feedback than are available

Research gap: no integrated telecom framework from target engineering to forecasting, benchmarking, and supervisory control under real telemetry constraints

Sorrentino et al. (2024); D'Aniello et al. (2018); Malafronte et al. (2021); Fan et al. (2021); Ji et al. (2017); Killian and Kozek (2016); Drgoňa et al. (2020); Xin et al. (2024)

Research aim and objectives

Aim: Develop and evaluate intelligent control strategies for cooling energy use in telecom central office environments using real monitoring data

Research tasks:

1. Review relevant literature
2. Prepare the control-oriented dataset
3. Develop the cooling-demand benchmark
4. Design context-aware control strategies
5. Evaluate control performance historically

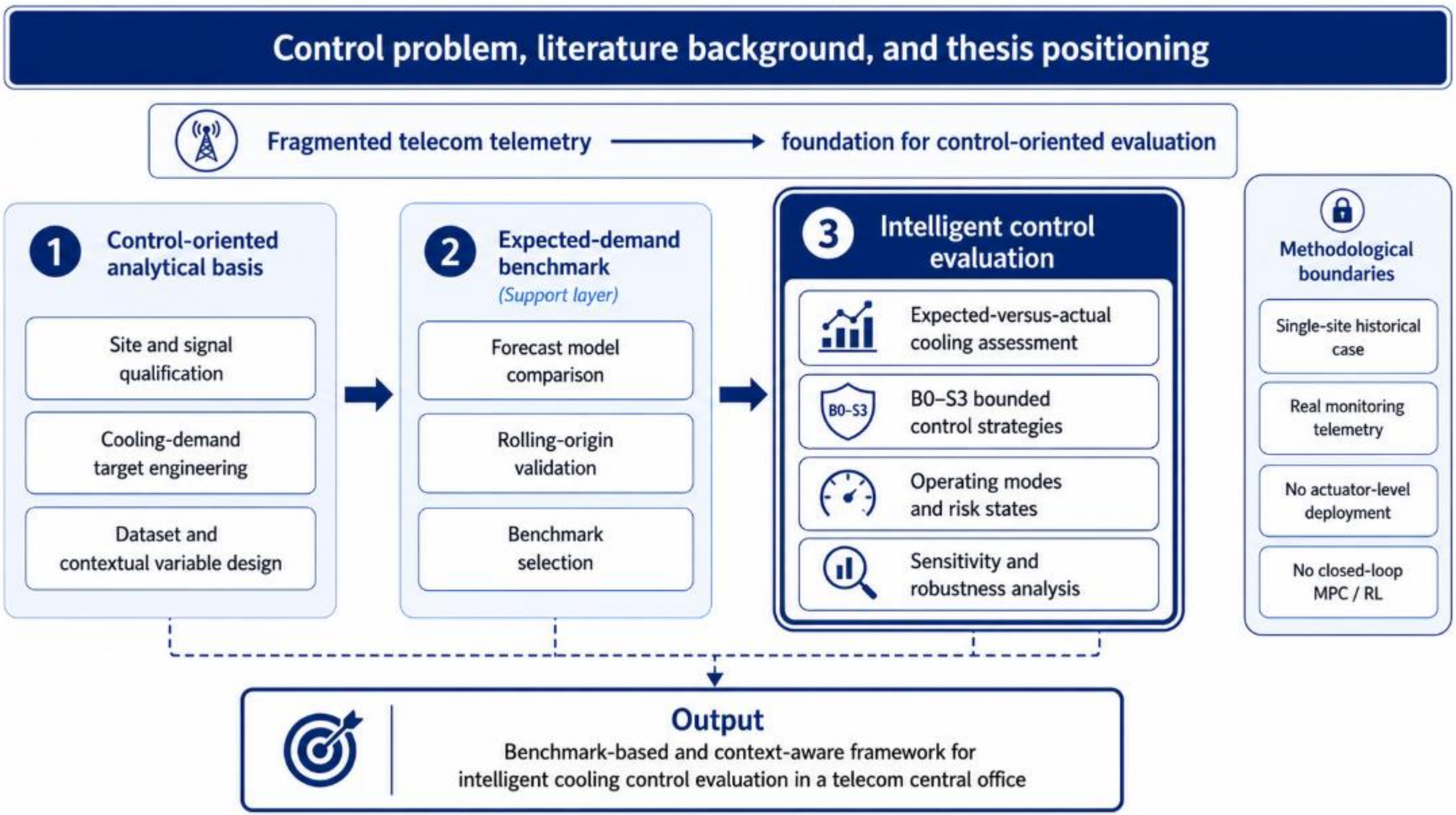
Research questions

1. To what extent can intelligent cooling control strategies identify opportunities to improve cooling energy use in telecom central office environments?
2. Which forecasting approach provides the most reliable estimate of expected cooling demand for intelligent cooling control?
3. Which operational and environmental variables are most relevant for constructing a control-oriented analytical basis for intelligent cooling control?

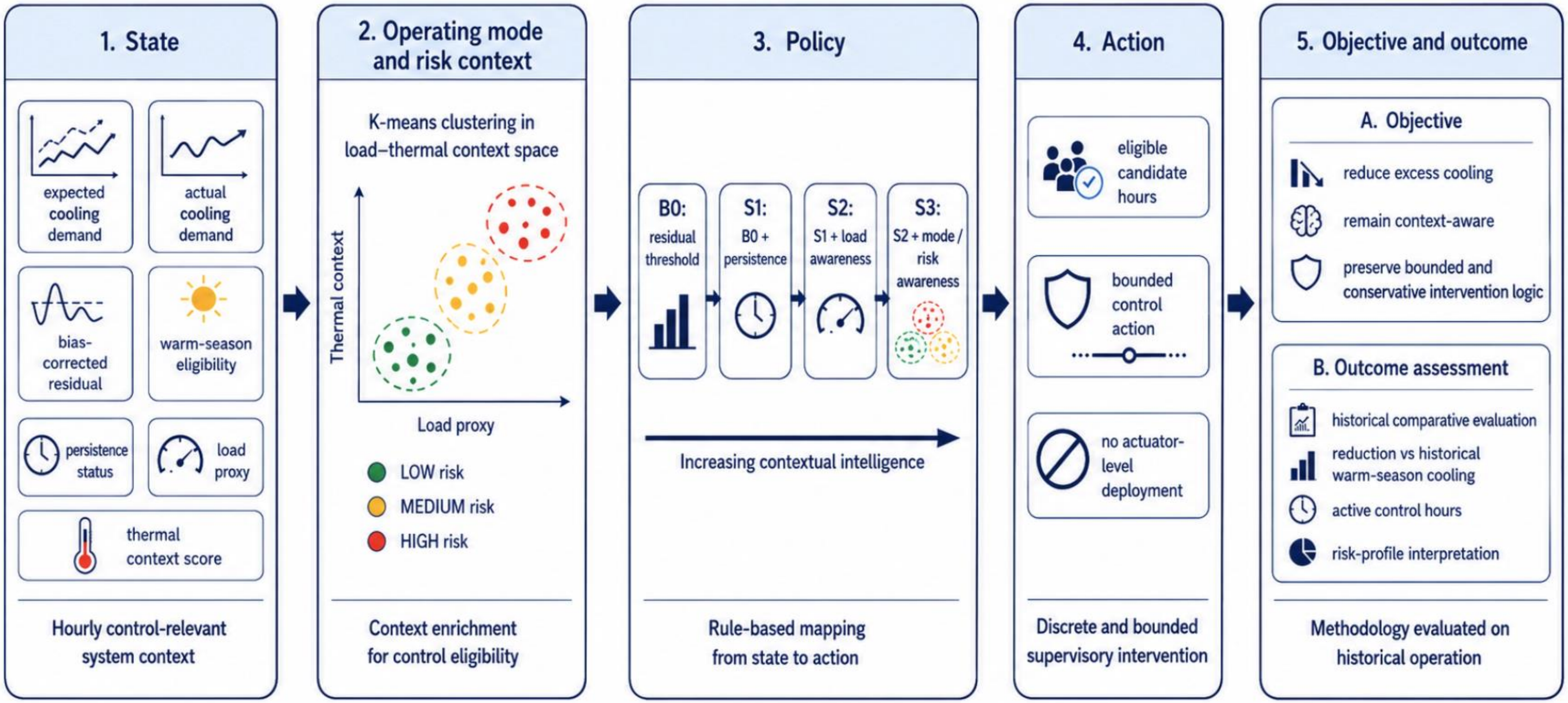
Object and subject of research

- **The object of the research** is cooling energy use in telecom central office technical infrastructure.
- **The subject of the research** is intelligent cooling control in telecom central office environments under real operational telemetry constraints

Analytical framework

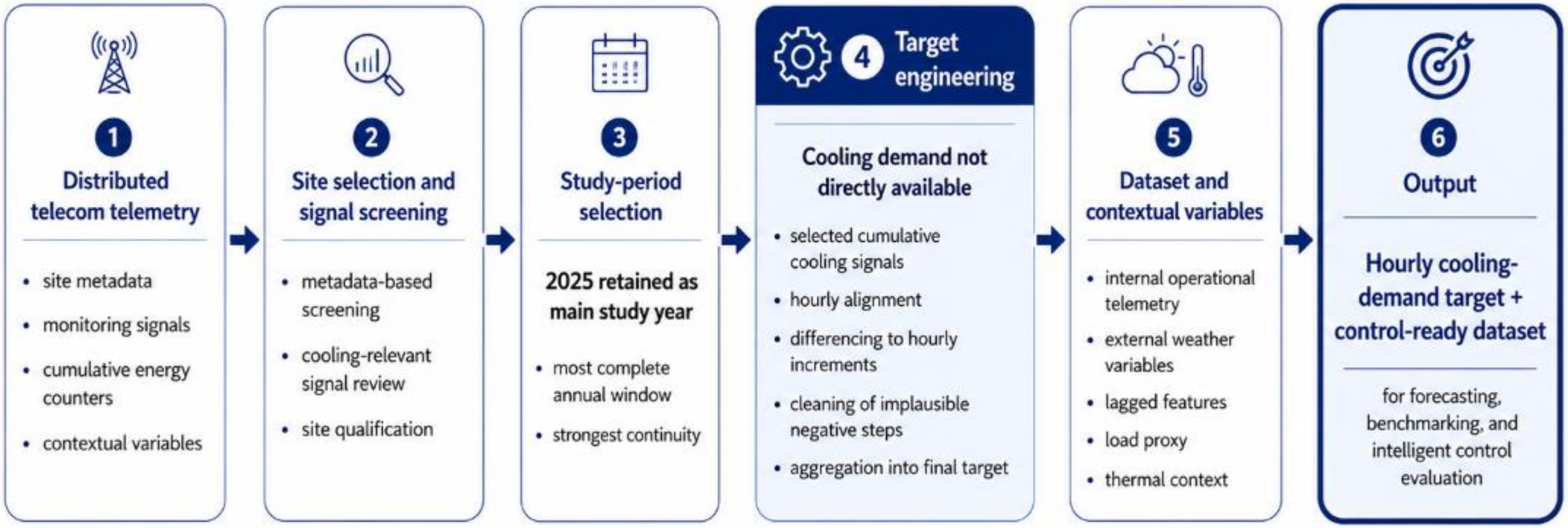


Intelligent control strategy



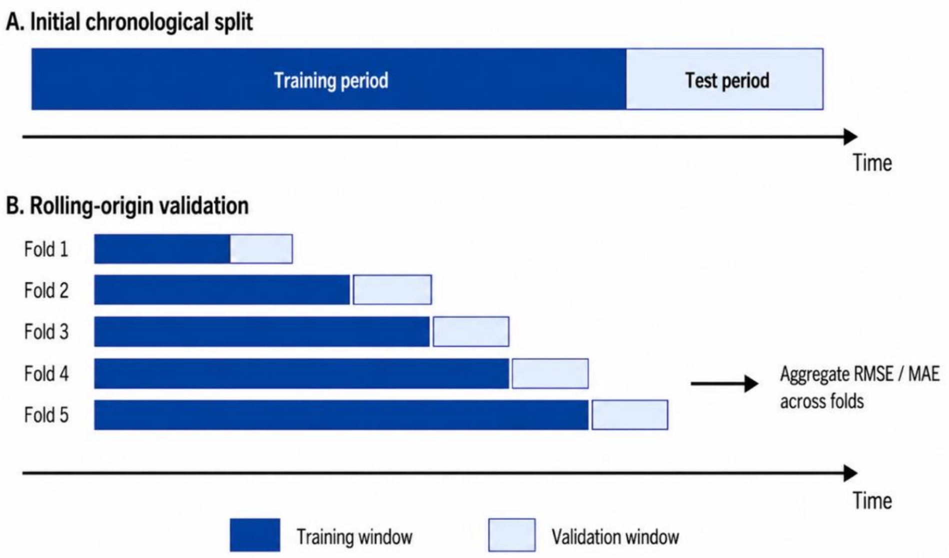
From telecom telemetry to cooling target

- Cooling demand was not directly available as a ready-made hourly variable
- Site screening, cumulative-signal processing, and contextual variable design produced a control-ready dataset



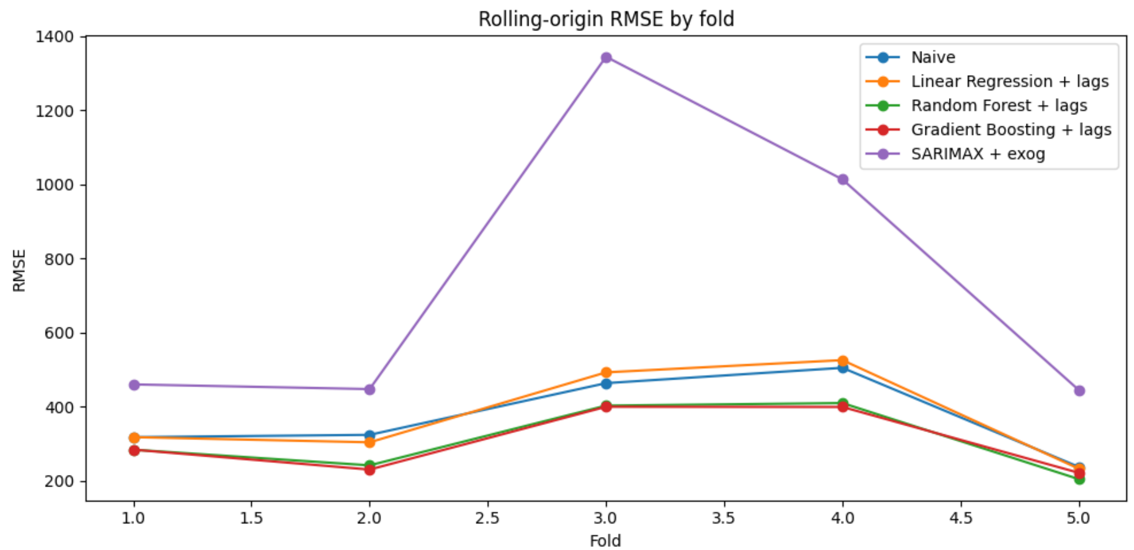
Forecasting and validation

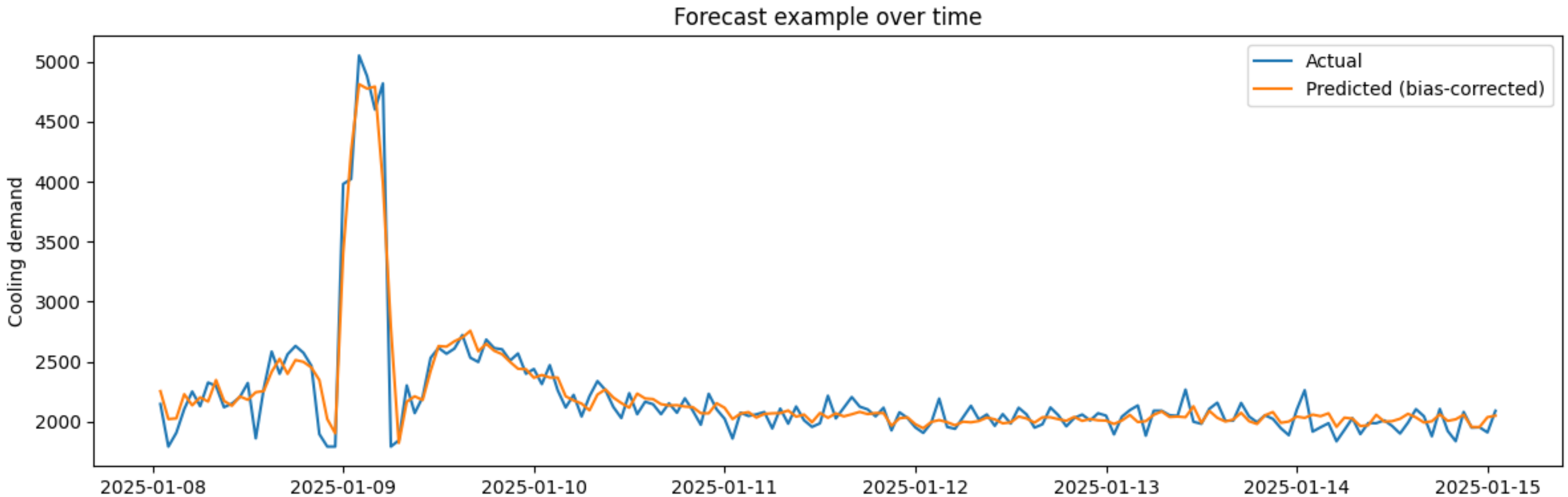
- Models compared: naive, linear regression, RF, GB, SARIMAX, LSTM
- Final lag set: **1, 2, 3, 6, 12, 24, 48, 168 hours**
- Metrics: **MAE, RMSE, R^2**
- Validation: **chronological split + rolling-origin**



Expected demand benchmark selection

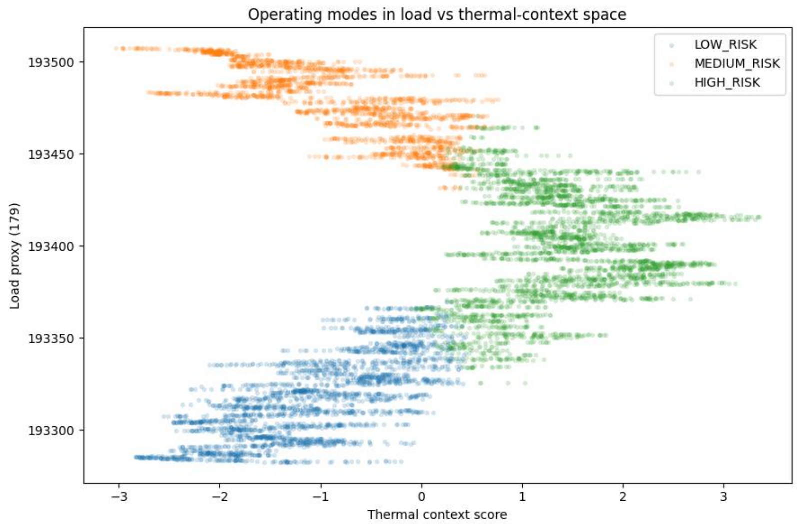
- **Gradient Boosting + lags**
- Final benchmark quality:
  - **MAE = 115.06**
  - **RMSE = 161.79**
  - **R^2 = 0.9855**

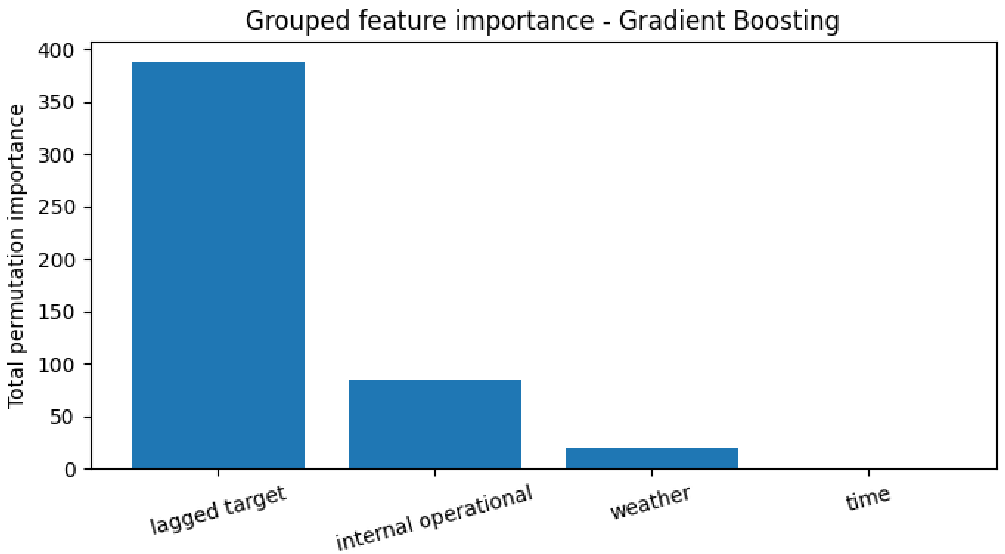




Most influential inputs in the analytical basis

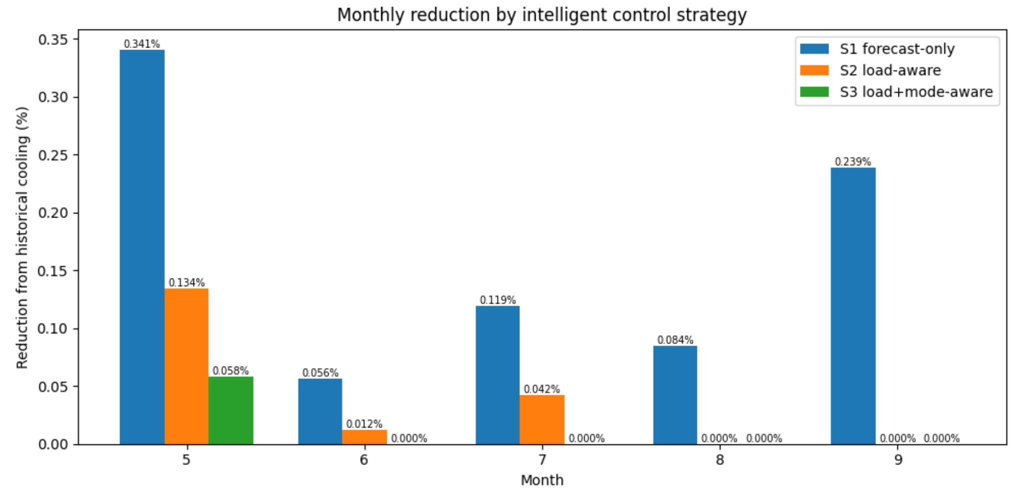
- Lagged history was the strongest predictor group
- Control-state design then added load proxy and thermal context for policy logic

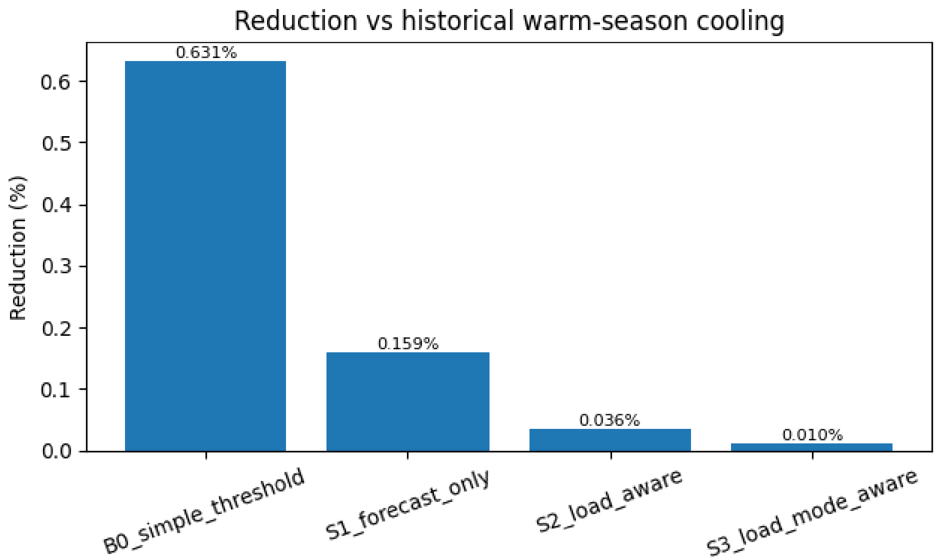


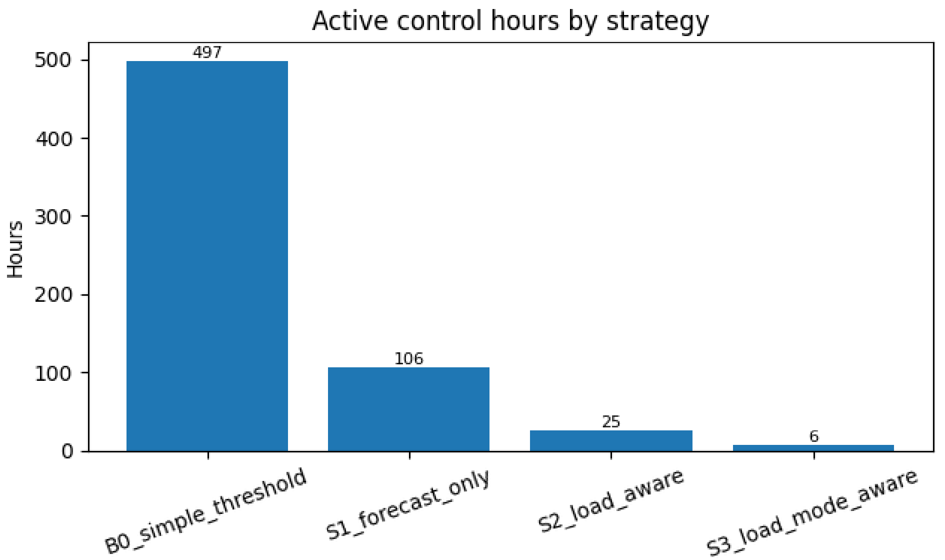


Control outcome and validation

- More context led to fewer active control hours
- B0 identified 497 hours, while S3 retained only 6
- Estimated reduction fell from 0.631% in B0 to 0.010% in S3

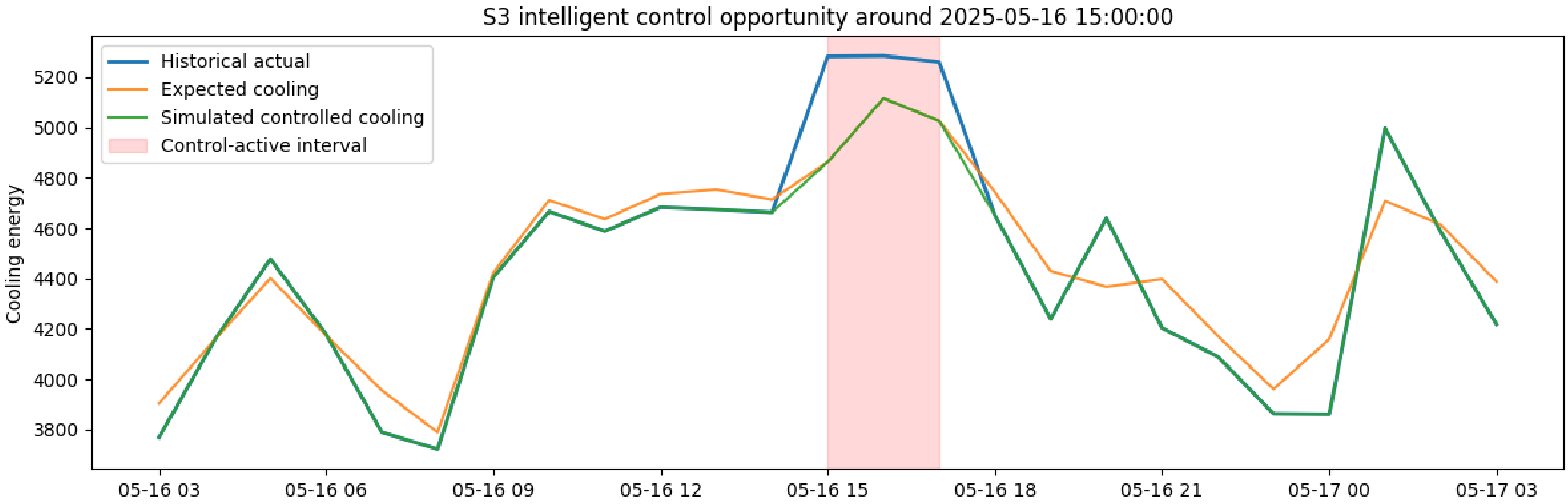






Example of control application

- Historical cooling exceeded the expected-demand benchmark
- Control was applied only when eligibility conditions were met
- Simulated cooling was reduced in a bounded way
- The 24h window makes persistence and activation context visible



Discussion and limitations

1. The method found small and selective control opportunities, not large general savings
2. The main value of the thesis is the framework which shows how control can be analysed with limited telecom data
3. The results are based on a benchmark model, not on direct measurement of full indoor thermal conditions
4. The study used one site and historical data only, so real-time testing is still needed

Conclusions

- The thesis achieved its aim by showing that intelligent cooling control can be analysed in a telecom central office under real telemetry constraints
- In the present case, intelligent control was formulated as a bounded and context-aware analytical framework rather than as direct live control
- All main tasks were completed: literature review, construction of the analytical basis, expected-demand benchmark development, control-strategy design, and historical evaluation
- The study produced a framework linking fragmented telemetry, expected-demand benchmarking, and context-aware intelligent control assessment

Conclusions

- **RQ1**: Intelligent cooling control identified a small but defensible set of control opportunities, which became much narrower when persistence, load context, and risk-state conditions were added
- **RQ2**: The most reliable estimate of expected cooling demand was provided by a lagged nonlinear tree-based benchmark, with Gradient Boosting selected as the final model
- **RQ3**: The analytical basis depended most on the engineered cooling target, lagged demand history, and the contextual control-state variables: weather context, load proxy, and thermal context
- The thesis supports a benchmark-based and context-aware approach to intelligent cooling control in telecom environments

Thank you, any questions?

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